

Week 8 - Convection and SALR: Worked Answers

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS:** selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY:** expected independent practice after the session.
- **EXTRA CHALLENGE:** optional problems for students who want additional depth or a harder application.

The categories are repeated here so the worked answers map directly onto the practice sheet. Use the solutions to check method, units and physical interpretation rather than only the final number. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

Use $\Gamma_d = 9.8 \text{ K km}^{-1}$ for the dry adiabatic lapse rate. Where a simple numerical value is required, use a representative saturated adiabatic lapse rate $\Gamma_s = 6.0 \text{ K km}^{-1}$; remember that the true SALR varies with temperature and pressure. For dry potential temperature use $\theta = T(p_0/p)^{R_d/c_p}$ with $R_d/c_p = 0.286$ and $p_0 = 1000 \text{ hPa}$.

YOU SHOULD TRY Question 1: Quantify the effect of latent heating on a rising parcel

Two parcels start at 20°C and rise 2.0 km . Parcel A remains unsaturated throughout and cools at the DALR. Parcel B is saturated throughout and, for this exercise, cools at 6.0 K km^{-1} .

1. Calculate the final temperature of each parcel.
2. By how much warmer is the saturated parcel after the ascent?
3. Identify the physical process responsible for the difference.
4. Give one reason why the true saturated adiabatic lapse rate is not a universal constant.

Worked answer

For the unsaturated parcel,

$$T_A = 20 - 9.8(2.0) = 0.4^\circ\text{C}.$$

For the saturated parcel,

$$T_B = 20 - 6.0(2.0) = 8.0^\circ\text{C}.$$

Thus Parcel B is 7.6 K warmer after the same ascent. Condensation in the saturated parcel releases latent heat, partly offsetting expansion cooling. The true SALR varies because the amount of vapour available to condense depends strongly on temperature and pressure; warm saturated air contains much more water vapour than cold saturated air.

IN CLASS Question 2: Lift a parcel through cloud base

A surface parcel has $T = 25^\circ\text{C}$ and $T_d = 17^\circ\text{C}$.

1. Estimate its LCL using $z_{LCL} = 125(T - T_d)$ m.
2. Estimate the parcel temperature at the LCL using the DALR.
3. If the parcel then rises to 3.0 km and cools at 6.0 K km^{-1} above the LCL, estimate its temperature at 3.0 km.

Worked answer

The temperature-dew-point difference is 8°C , so

$$z_{LCL} = 125 \times 8 = 1000 \text{ m} = 1.0 \text{ km}.$$

Below cloud base,

$$T_{LCL} = 25 - 9.8(1.0) = 15.2^\circ\text{C}.$$

The parcel then rises a further 2.0 km while saturated:

$$T_{3km} = 15.2 - 6.0(2.0) = 3.2^\circ\text{C}.$$

IN CLASS Question 3: Classifying atmospheric stability

For each environmental lapse rate below, classify the atmosphere as absolutely stable, conditionally unstable or absolutely unstable. Use $\Gamma_s = 6.0$ and $\Gamma_d = 9.8 \text{ K km}^{-1}$.

1. $\Gamma_{env} = 4.0 \text{ K km}^{-1}$
2. $\Gamma_{env} = 7.0 \text{ K km}^{-1}$
3. $\Gamma_{env} = 11.0 \text{ K km}^{-1}$

For each case, state what happens to an air parcel after a small upward displacement.

Worked answer

1. $4.0 < 6.0$: absolutely stable. Both dry and saturated lifted parcels cool faster than the environment and tend to become colder/denser, so the displacement is opposed.
2. $6.0 < 7.0 < 9.8$: conditionally unstable. An unsaturated parcel is stable, but a saturated parcel can cool more slowly than the environment and become positively buoyant.
3. $11.0 > 9.8$: absolutely unstable. Even an unsaturated lifted parcel remains warmer than the environment and tends to continue rising.

YOU SHOULD TRY Question 4: Potential temperature

An unsaturated air parcel has $T = 280 \text{ K}$ at 800 hPa .

1. Calculate its dry potential temperature.
2. A second parcel at the same pressure has $\theta = 304 \text{ K}$. Which parcel is warmer at 800 hPa ?
3. State why potential temperature is useful for dry adiabatic motion.

Worked answer

$$\theta = 280 \left(\frac{1000}{800} \right)^{0.286} = 298.4 \text{ K}.$$

At the same pressure, the parcel with the greater potential temperature also has the greater actual temperature, so the second parcel is warmer. Potential temperature is conserved for dry adiabatic motion, so it allows air parcels at different pressures to be compared on a common reference pressure.

IN CLASS Question 5: Find an inversion and diagnose its consequences

A morning temperature profile is

Height (m)	0	200	500	800	1200
Temperature (°C)	5	3	7	5	2

1. Between which two levels is there a temperature inversion?
2. Calculate the temperature change across that inversion layer.
3. Would vertical mixing through the inversion be encouraged or suppressed?
4. A city lies below the inversion under light winds. Predict the effect on near-surface pollutant concentration and explain why.

Worked answer

Temperature increases from 3°C at 200 m to 7°C at 500 m, so the inversion lies across that layer and the temperature change is +4 K.

An inversion is strongly stable: an air parcel displaced upward tends to become colder and denser than the warmer environment, so vertical mixing is suppressed. Under light winds, pollutants emitted near the surface are then trapped in a shallow layer and can accumulate to higher concentrations.

EXTRA CHALLENGE Question 6: A simple Föhn-wind calculation

Air approaches a mountain at sea level with temperature 20°C. Its LCL is at 1.0 km and the summit is at 3.0 km. Assume it cools at the DALR below the LCL and at 6.0 K km⁻¹ while saturated above it. On the lee side, assume condensed water has precipitated out and the descending air warms at the DALR all the way to sea level.

1. Calculate the temperature at the summit.
2. Calculate the temperature on the lee side at sea level.
3. By how many degrees is the lee-side air warmer than the original windward surface air?
4. Explain why the descending air is relatively dry.

Worked answer

From sea level to the 1 km LCL,

$$T = 20 - 9.8 = 10.2^\circ\text{C}.$$

From 1 to 3 km the saturated parcel cools by 12 K, so

$$T_{\text{summit}} = 10.2 - 12 = -1.8^\circ\text{C}.$$

Descending 3 km on the lee side at the DALR gives

$$T_{\text{lee}} = -1.8 + 9.8(3) = 27.6^\circ\text{C}.$$

The lee-side air is therefore 7.6 K warmer than the original surface air. Moisture was removed by condensation and precipitation during the windward ascent; the descending air is unsaturated and warms rapidly by compression.

EXTRA CHALLENGE Question 7: Identify the lifting mechanism from the weather situation

For each situation, name the main mechanism that initiates ascent.

1. Strong afternoon heating produces rising thermals over a dark land surface.
2. A moist westerly wind encounters the Pennines.
3. A cold front advances beneath warm air.
4. Surface winds from several directions meet along a narrow boundary.

For one of the four cases, explain why ascent does not automatically guarantee deep convection.

Worked answer

The mechanisms are: (1) surface buoyancy/heating, (2) orographic lifting, (3) frontal lifting, and (4) low-level convergence.

Ascent alone does not guarantee deep convection because the environmental stability and moisture profile also matter. A lifted parcel may remain negatively buoyant, may not reach saturation, or may encounter a stable inversion before reaching its level of free convection.

YOU SHOULD TRY Question 8: Use stability to predict cloud form

Two saturated layers have the following environmental lapse rates:

- Layer A: 3.0 K km^{-1}
- Layer B: 8.0 K km^{-1}

Use $\Gamma_s = 6.0 \text{ K km}^{-1}$.

1. Which layer is stable with respect to saturated vertical displacement?
2. Which layer is unstable with respect to saturated vertical displacement?
3. If cloud forms in each, which layer is more likely to favour horizontally extensive layered cloud, and which is more likely to favour vertical cumulus development? Explain from parcel-environment temperature differences.

Worked answer

Layer A has $\Gamma_{env} < \Gamma_s$, so it is stable for saturated parcels: a lifted saturated parcel cools faster than the environment and becomes negatively buoyant. Layer B has $\Gamma_{env} > \Gamma_s$, so a saturated lifted parcel can become warmer than its environment and continue rising.

Layer A therefore favours shallow, horizontally extensive cloud such as stratus/altostratus when saturation is present. Layer B is more favourable for vertically developing cumulus or cumulonimbus, provided sufficient moisture and a lifting trigger are available.

EXTRA CHALLENGE Question 9: Parcel versus environment through a conditionally unstable layer

At 1.0 km, a saturated parcel and its environment are both at 12°C . Between 1 and 3 km the environment cools at 7.5 K km^{-1} while the parcel cools at 6.0 K km^{-1} .

1. Calculate parcel and environmental temperatures at 3 km.
2. Which is warmer, and by how much?
3. What does this imply about the parcel's buoyancy tendency?

Worked answer

Over 2 km the parcel cools by 12 K, so $T_p = 0^\circ\text{C}$. The environment cools by 15 K, so $T_e = -3^\circ\text{C}$. The parcel is therefore 3 K warmer than its environment and tends to be positively buoyant, consistent with saturated instability in this layer.